

postgraduates in technology. "All IITs, NITs, BITS, Anna University, VJTI Mumbai and PSG Coimbatore offer good training on this subject," shares Teresa Mishra, general manager-talent acquisition, Wipro Technologies.

IIT Bombay offers a dual-degree programme combining B.Tech in electrical engineering and M.Tech in microelectronics. The fee structure for this five-year course is the same as for all other IIT courses, i.e., Rs 23,000-24,000 per semester. Some of the subjects taught as part of this programme are VLSI design, analogue design, embedded systems, FPGAs, physics of transistors, systems-on-chips, VLSI testing and embedded systems.

At IIT Kharagpur, students can learn digital design, analogue design and VLSI engineering. They are taught analogue VLSI circuit, digital VLSI circuit, semiconductor design modeling, architectural design of VLSI circuit and VLSI technology for fabrication.

Money matters

The private sector offers lofty pay packages to candidates having a suitable exposure in the form of proper industrial training or systematic industry-defined project work. Typically, the starting salary for a fresher (diploma or degree holder) is Rs 400,000 upwards per annum, with the variation depending on the nature of the organisation and competency level of the candidate. Design engineers are the most sought after because of the industry's emphasis on continuous new product development, miniaturisation and innovation in integration.

Apart from multinationals, the government also hires chip designers for space and defence research — Anindya Sundar Dhar, associate professor and faculty advisor, microelectronics and VLSI design, IIT Kharagpur

"We work closely with many universities in the country and help them with setting up of labs so that students get opportunities to work on practical aspects as part of their coursework"

— Sanjay Bhan, director-HR, Texas Instruments India



Sanjay Bhan shares, "A career in the semiconductor industry is highly rewarding both for new college graduates and experienced people. Today, in India, we have many local as well as international companies that are doing highly innovative work that is valued globally."

"The salary across the market is very good for experienced candidates. It is also equally rewarding for freshers in terms of exposure and growth," quips Mishra.

"The annual salary for postgraduates may range from Rs 600,000 to 800,000 and for Ph.D holders from Rs 1.1 to 1.8 million," informs Prof. Rao.

Anindya Sundar Dhar, associate professor and faculty advisor, microelectronics and VLSI design, IIT Kharagpur, adds, "The salary for an M.Tech from IIT starts at Rs 1 million per annum. Apart from multinationals, the government also hires chip designers for space and defence research. They look for sound domain knowledge."

Skills in demand

Intel, IBM, Texas Instruments, Analog Devices, Cadence and Synopsis are some of the big recruiters in India. Companies want students who have designed a chip as part of their curriculum

and also tested it in silicon.

Prof. Rao shares, "Our students not only design the chip but also send it for fabrication to companies that provide this service and get it tested. They get exposure in design tools and silicon implementation."

Texas Instruments looks for candidates with an understanding of the fundamentals of electronics. "There are many branches in electronics today such as VLSI, DSP, electronic communications, power electronics and embedded systems. We expect the prospective candidate to have studied at least one course in these branches. Although we do provide job-specific training, it is expected that the candidate has technical knowledge required to absorb the training. For example, a training session on VLSI may require the student to know about MOS transistor operation, CMOS circuits, logic gates and flip-flops, operational amplifiers, feedback amplifiers, poles and zeroes," Bhan states.

Wipro Technologies, on the other hand, looks for candidates with specific knowledge on digital system design concepts, timing concept and semiconductor physics.

"Candidates looking for careers in this field should also be well versed with languages like VHDL, Verilog, Specman and System Verilog. Freshers are given an intensive training for three months," informs Mishra.

Sunil Pathak, director-HR, Cadence Design Systems, says, "Cadence expects the candidates to have a good knowledge of the VLSI design flow. From the technical perspective, they

